

8.3 POPULATION INTERACTIONS

8.3.1 Inter-specific Interactions

8.3.2 Predation

8.3.3 Competition

8.3.4 Parasitism

8.3.5 Commensalism

8.3.6 Mutualism

Can you think of any natural habitat on earth that is inhabited just by a single species? There is no such habitat and such a situation is unimaginable. For any species, the minimal requirement is one more species on which it can feed. Even a plant species, which makes its own food, cannot survive alone; it needs soil microbes to break down the organic matter in soil and return the inorganic nutrients for absorption. And

then, how will the plant manage pollination without an animal agent? It is obvious that in nature, animals, plants and microbes do not and cannot live in isolation but interact in various ways to form a **biological community**. Even in minimal communities, many interactive linkages exist, although all may not be readily visible.

8.3.1 Inter-specific Interactions

Inter-specific interactions arise from the interaction of populations of two different species. They could be beneficial, detrimental or neutral (neither harmful nor beneficial) to one of the species or both. Assigning a '+' sign for beneficial interaction, '-' sign for detrimental and '0' for neutral interaction, let us look at all the possible outcomes of inter-specific interactions.

The interactions between species are grouped into six types. They are mutualism, commensalism, parasitism and amensalism. Both the species benefit in **mutualism** and both lose in **competition** in their interactions with each other. The interaction where one species is benefitted and the other is neither benefitted nor harmed is called **commensalism**. In **amensalism** on the other hand one species is harmed whereas the other is unaffected. In both **parasitism** and **predation** only one species benefits (parasite and predator, respectively) and the interaction is detrimental to the other species (**host** and **prey**, respectively). Predation, parasitism and commensalisms share a common characteristic - the interacting species live closely together.

Table 8.1: Population Interactions-Types

Name of Interaction	Species A	Species B
Mutualism	+	+
Competition	-	-
Predation	+	-
Parasitism	+	-
Commensalism	+	0
Amensalism	-	0

[¶] Predation is not an association (it is a feeding strategy). It is an interaction between two different species. The predator gets benefit at the cost of the prey.

8.3.2 Predation

What would happen to all the energy fixed by autotrophic organisms if the community has no animals to eat the plants? You can consider predation as nature's way of transferring the energy fixed by plants to higher trophic levels. When we think of predator and prey, most probably it is the tiger and the deer that readily come to our mind, but a sparrow eating any seed is also a type of predator (a seed predator-also called **granivore**). Although animals eating plants are categorized separately as herbivores, they are, in a broad ecological context, not very different from predators.

Besides acting as 'conduits' / 'pipelines' for energy transfer across trophic levels, predators play other important roles. They keep the prey populations under control. In the absence of predators, the prey species could achieve very high population densities and cause instability in the ecosystem. Predators have different types of functions to play in nature. They include:

A. Predator as a biological control agent

The prickly pear cactus introduced into Australia in the early 1920s caused havoc by spreading rapidly into millions of hectares of rangeland (vast natural grass lands). Finally, the invasive cactus was brought under control only after a cactus-feeding predator (a moth) was introduced into the country. **Biological control** methods adopted in agricultural pest control are based on the ability of the predators to regulate prey populations.

B. Predators maintain 'species diversity'

Predators also help in maintaining species diversity in a community, by reducing the intensity of competition among competing prey species. In the rocky intertidal communities of the American Pacific Coast, the starfish *Pisaster* is an important predator. In a field experiment, when all the starfish were removed from an enclosed intertidal area, more than 10 species of invertebrates became extinct within a year in that experimental area, because of increased **inter - specific competition**.

C. Predators are prudent (practical) pertaining to preys

If a predator is too efficient and overexploits its prey, then the prey might become extinct and following it, the predator will also become extinct due to lack of food. This is the reason why predators in nature are 'prudent'.

Prey species have evolved various defenses to lessen the impact of predation. They include :

- a) **Preys fool (deceive) or avoid their predators**: Some species of insects and frogs are cryptically-coloured (**camouflaged**) to avoid being detected easily by the predator. Some are poisonous and therefore avoided by the predators.
- b) **Preys defend by becoming distasteful to predators**: The **Monarch butterfly** is highly distasteful to its predator (bird) because of a special chemical present in its

body. Interestingly, the butterfly acquires this chemical during its **caterpillar stage** by feeding on a **poisonous weed**.

- c) Plants too have their defensive mechanisms:** For plants, herbivores are the predators. Nearly 25 per cent of all insects are known to be **phytophagous** (feeding on plant sap and other parts of plants). The problem is particularly severe for plants because, unlike animals, they cannot escape from their predators. Plants therefore have evolved a variety of morphological and chemical defences against herbivores.
- i) **Thorns** (Acacia, Cactus, etc.,) are the most common morphological means of defense. Many plants produce and store chemicals that make the herbivore sick when they are eaten, inhibit feeding or digestion, disrupt its reproduction or even kill it.
 - ii) You must have seen the weed Calotropis growing in abandoned fields. The plant produces highly poisonous **cardiac glycosides** and that is why you never see any cattle or goats browsing on this plant.
 - iii) A wide variety of chemical substances that we extract from plants on a commercial scale (**nicotine, caffeine, quinine, strychnine, opium**, etc.,) are produced by them actually as defences against grazers and browsers.

8.3.3 Competition

When Darwin spoke of the struggle for existence and survival of the fittest in nature, he was convinced that **interspecific competition** is a 'potent force' in the process of organic evolution, involving Natural Selection. It is generally believed that competition occurs when closely related species compete for the same resources that are limited, but this is not entirely true.

I. Competition among Unrelated Species

Firstly, unrelated species could also compete for the same resource (Interspecific competition). For instance, in some **shallow** South American lakes visiting **flamingos** and resident **fishes** compete for their common food, the suspended **zooplankton** in the shallow waters. Secondly, resources need not be limiting for competition to occur. In **interference competition**, the feeding efficiency of one species might be reduced due to the interfering and inhibitory presence of the other species, even if resources (food and space) are abundant. Therefore, competition is best defined as a process in which the fitness of one species (measured in terms of its '**r**'- the intrinsic rate of increase) is significantly lower in the presence of another species.

II. Competitive exclusion

It is relatively easy to demonstrate in laboratory experiments, as **Gause** and other experimental ecologists did. When the resources are limited, the competitively superior species will eventually eliminate the other species e.g. the Abingdon tortoise in Galapagos

Islands became extinct within a decade after goats were introduced on the island, actually due to the greater browsing efficiency of the goats.

III. Competitive release

Another evidence for the occurrence of competition in nature comes from what is called '**competitive release**'. Competitive release occurs when one of the two competing species is removed from an area, thereby releasing the remaining species from one of the factors that limited its population size. A species, whose distribution is restricted to a small geographical area because of the presence of a competitively superior species, is found to expand its distributional range dramatically when the **competing species** is experimentally removed. This is due to the phenomenon called '**competitive release**'. **Connell's** 'field experiments' showed that, on the rocky sea coasts of Scotland, the larger and competitively superior barnacle *Balanus* dominates the intertidal area, and excludes the smaller barnacle *Chthamalus* from that zone. When the dominant one is experimentally removed, the populations of the smaller ones increased. In general, herbivores and plants appear to be more adversely affected by competition than the carnivores (Ref: NCERT text book).

IV. Coexistence, rather than exclusion

Gause's principle of 'Competitive Exclusion' states that two closely related species competing for the same resources cannot co-exist indefinitely and the **competitively inferior one** will be eliminated in due course of time. This may be true if resources are limiting, but not otherwise.

More recent studies point out that species facing competition might evolve mechanisms that promote **co-existence** rather than competitive exclusion. One such mechanism is '**resource partitioning**'. If two species compete for the same resource, they could avoid competition by choosing, for instance, different times for feeding or different foraging (food collecting) patterns. **MacArthur** showed that five closely related species of warblers (a kind of birds) living on the same tree were able to avoid competition and co-exist due to behavioural differences in their foraging activities.

8.3.4 Parasitism

Considering that the parasitic mode of life ensures free 'lodging' and 'meals', it is not surprising that parasitism has evolved in so many taxonomic groups from plants to higher vertebrates. Many parasites have evolved to be **host-specific** (they can parasitize only a specific species of host) in such a way that both host and the parasite tend to **co-evolve**; that is, if the host evolves special mechanisms for rejecting or resisting the parasite, the parasite has to evolve mechanisms to 'counteract' and 'neutralize' them, in order to continue successful parasitic relationship with the same host species. In order to lead successful parasitic life, parasites evolved special adaptations, such as:

- a. Loss of sense organs (which are not necessary for most parasites).
- b. Presence of adhesive organs such as suckers, hooks to cling on to the host's body parts.
- c. Loss of digestive system and presence of high reproductive capacity.
- d. The life cycles of parasites are often **complex**, involving one or two intermediate hosts or vectors to facilitate parasitisation of their primary hosts.

e.g.-1: The human liver fluke depends on two **intermediate** (secondary) **hosts** (a **snail** and a **fish**) to complete its life cycle.

e.g.-2: The malaria parasite needs a vector (mosquito) to spread to other hosts.

Majority of the parasites harm the host; they may reduce the survival, growth and reproduction of the host and reduce its population density. They might render the host more vulnerable to predation by making it physically weak.

The Types of Parasites

- I. **Ectoparasites:** These parasites feed on host organism while remaining outside the body of the host. The most familiar examples of this group are the **lice** on humans and **ticks** on dogs. Many marine fish are infested with ectoparasitic copepods.

Plant examples: *Cuscuta*, a parasitic plant that is commonly found growing on hedge plants, has lost its chlorophyll and leaves in the course of evolution. It derives its nutrition from the host plant which it parasitises. (Ref: NCERT Text Book)

- II. **Endoparasites:** Endoparasites are those that live inside the host's body at different sites (liver, kidney, lungs, red blood cells, etc.). The life cycles of endoparasites are more complex because of their extreme specialization. Their morphological and anatomical features are greatly simplified while emphasizing (stressing on) their reproductive potential.
- III. **Brood parasitism:** Certain birds show a special type of parasitism, in which the 'parasitic bird' lays its eggs in the nest of its host and lets / allows the host to incubate them. During the course of evolution, the eggs of the parasitic bird have evolved to resemble the host's egg in size and colour to reduce the chances of the host bird detecting the foreign eggs and ejecting them from the nest. This is treated as a special type of parasitism by some

Try to follow the movements of the cuckoo (koel) and the crow in your neighborhood park during the breeding season (spring to summer) and watch brood parasitism in action (NCERT Text Book).

8.3.5 Commensalism

This is the interaction in which one species benefits and the other is neither harmed nor benefited. Barnacles growing on the back of a whale benefit while the whale derives no noticeable benefit.

The ‘cattle egret’ (a kind of bird) and ‘grazing cattle’ living in close association, is a good example of commensalism. The egrets always forage close to where the cattle are grazing because the cattle, as they move, stir up and drive out insects, from the vegetation thus helping the egrets to find and catch the insects. Another example of commensalism is the interaction between **sea anemone** that has tentacles with **stinging cells** and the ‘**clown fish**’ that lives among them. The fish gets protection from predators, as the predator stays away from the stinging tentacles of the anemone. The anemone does not derive any visible benefit by ‘hosting’ the clown fish.

Plant example :An orchid growing as an epiphyte on a mango branch, gets the benefit of exposure to light, while the mango tree does not derive any noticeable benefit (Ref: NCERT Text Book)

8.3.6 Mutualism

This type of interaction benefits both the interacting species.

The most common examples of mutualism are found in plant-animal relationships. Plants need the help of animals for **pollinating** their flowers and **dispersing** their seeds. Animals obviously have to be paid ‘**fees**’ for the services that plants derive from them. Plants offer rewards in the form of **pollen** and **nectar** for pollinators and juicy and nutritious **fruits** for seed dispersing animals.

Now you can see why plant-animal interactions often involve **co-evolution** of the mutualists, that is, the evolutions of the flower and its pollinator species are tightly linked with one another.



Fig. 8.8 Commensalism

Plant Examples: Lichens represent an intimate mutualistic relationship between a fungus and photosynthesising algae or cyanobacteria. Similarly, the **mycorrhizae** are associations between fungi and the roots of higher plants. The fungi help the plant in the absorption of essential nutrients from the soil while the plant in turn provides the fungi with energy-yielding carbohydrates (Ref: NCERT Text Book).



Fig. 8.9 Mutual relationship between fig tree and wasp (a) Fig flower is pollinated by wasp; (b) Wasp laying eggs in a fig fruit

In many species of fig trees, there is a one-to-one relationship with the pollinator species of **wasp**. It means that a given fig species can be pollinated only by its '**partner**' **wasp** species and no other species. The female wasp uses the fruit not only as a site for oviposition (**egg-laying site**), but also uses the developing seeds within the fruit for nourishing its larvae. The wasp pollinates the flowers of the fig plant while searching for suitable egg-laying sites.

In return for the favour of pollination the fig offers the wasp some of its developing seeds, as food for the developing wasp larvae.



Fig. 8.10 Showing bee a pollinator on orchid flower

Orchids show a high degree of diversity of floral patterns. Many of them have evolved to attract the right pollinator insect (bees and bumblebees) and ensure pollination by it. Not all orchids offer rewards. The Mediterranean orchid *Ophrys* employs '**sexual deceit**' to get pollination done by a specific species of bee. One petal of its flower resembles the female of the bee in size, colour and markings. The male bee is attracted to the 'petal' as it presumes it to be the female, '**pseudocopulates**' with the flower, and during that process, it is dusted with pollen from the flower. When the same bee 'pseudocopulates' with another flower, it transfers the pollen to it. Here you can see how **co-evolution** operates in nature. If the female bee's colour patterns change even slightly, for any reason in the course of evolution, pollination success will be reduced, unless the orchid flower **co-evolves** and alters the resemblance of its petal to the changed female bee.

8.4. ECOSYSTEMS AND THEIR COMPONENTS

- 8.4.1 The Natural Ecosystems
- 8.4.2 Aquatic Ecosystems
- 8.4.3 Lake Ecosystem
- 8.4.4 Terrestrial Ecosystems
- 8.4.5 Artificial Ecosystems

An 'ecosystem' is a **functional unit of nature**, where living organisms interact among themselves and also with the surrounding physical environment. Ecosystem varies greatly in size from a small pond to a large forest or a sea. Many ecologists regard the entire **biosphere** as a 'global ecosystem', as a composite of all local ecosystems on Earth. Since this

system is too big and complex to be studied at one time, it is convenient to divide it into two basic categories, namely natural and artificial. The natural ecosystems include **aquatic ecosystems** of water and **terrestrial ecosystems** of the land. Both types of natural and artificial ecosystems have several subdivisions.

8.4.1 The Natural Ecosystems

These are naturally occurring ecosystems and there is no role of humans in the formation of such types of ecosystems. These are categorized mainly into two types - aquatic and terrestrial ecosystems.

8.4.2 Aquatic Ecosystems

Based on the **salinity** of water, three types of aquatic ecosystems are identified- marine, freshwater, and estuarine.

- i. **The Marine Ecosystem:** It is the largest of all the aquatic ecosystems. It is the most stable ecosystem
- ii. **Estuarine Ecosystem:** Estuary is the zone where river joins the sea. Sea water ascends up into the river twice a day (effect of high tides and low tides). The salinity of water in an estuary also depends on the seasons. During the rainy season out flow of river water makes the estuary less saline and the opposite occurs during the summer. Estuarine organisms are capable of withstanding the '**fluctuations**' in salinity.
- iii. **The Freshwater Ecosystem:** The freshwater ecosystem is the smallest aquatic ecosystem. It includes rivers, lakes, ponds, etc., It is divided into two groups- the **lentic** and **lotic**. The still water bodies like ponds, lakes, reservoirs, etc., fall under the category of **lentic ecosystems**, whereas, streams, rivers and flowing water bodies are called **lotic ecosystems**. The communities of the above two types are called lentic and lotic communities respectively. The study of freshwater ecosystem is called as **limnology**.

8.4.3 Lake Ecosystem

To understand the fundamentals of an aquatic ecosystem, let us take a 'lake' as an example. This is fairly a self-sustainable unit and rather a simple example that explains even the complex interactions that exist in an aquatic ecosystem.

Lakes are large inland water bodies containing standing/still water (Recall: Lentic community). They are deeper than ponds (pond is not an ideal example as it is very shallow). Most lakes contain water throughout the year. In deep lakes, light cannot penetrate more than 200 meters, in depth. They are vertically stratified in relation to light intensity, temperature, pressure, etc., Deep water lakes contain three distinct zones namely, i) littoral zone, ii) limnetic zone, and iii) profundal zone.

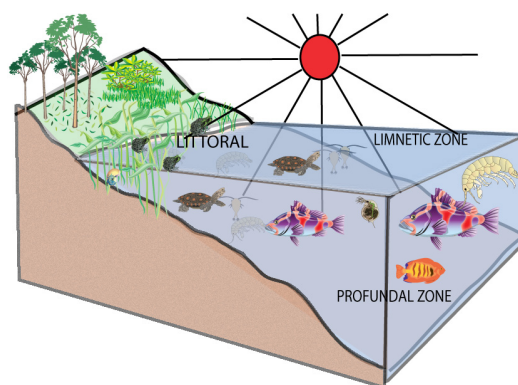


Fig. 8.11 Lake Ecosystem